

SOLUTIONS OF THE DIRAC EQUATION FOR GENERAL INSTANTON SOLUTIONS

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The zero mode normalisable solutions of the Dirac equation in the presence of a general self-dual gauge field of instanton number k for the groups $SU(n)$ or $Sp(r)$ are found. The method uses the explicit Green functions constructed by Corrigan et al. and gives the full complement of k -independent solutions and also determines their normalisations.

1. Introduction

It is a basic statement of the index theorem that in the context of a self dual gauge field of instanton number, or Pontryagin index, k the Euclidean Dirac equation has zero eigenvalue normalisable solutions and that the difference between the number of such solutions with negative and positive chirality (or helicity) is equal to k , (if the Dirac operator belongs to the fundamental representation of the gauge group) [1,2]. For the case of $SU(2)$ the well known 't Hooft solution [3] gives self dual gauge fields for arbitrary k and the corresponding solutions of the Dirac equation have also been explicitly found [2,4]. The zero mode eigenfunctions are of some importance in determining the structure of the functional integral for gauge theories [5].

However, the 't Hooft solution is not completely general, it does not exhaust the full allowed number of parameters, $8k - 3$, for a k -instanton solution [1,6]. Recently, general, and also complete, self-dual solutions for arbitrary gauge groups have been found in terms of matrices, linear in spatial variables, which satisfy some quadratic constraints and possible reality conditions [7,8]. The formalism has been spelled out in more physical terms by Corrigan et al. [9], who have further shown how the scalar Green function in the presence of a self-dual background field can be simply constructed. The elegance of the final result confirms the naturalness of this approach. From the scalar Green function it is then straightforward to construct the spinor Green's function for the Dirac operator [10]. Since this has zero eigenfunctions the completeness relation is modified by their explicit presence and so this allows their

determination given the basic Green function of Corrigan et al. [9].

This is the task undertaken in this paper, the calculation being given in sect. 3. Before this, in sect. 2 we establish the required notation, while in sect. 4 we reproduce known results corresponding to the 't Hooft case.

2. Formalism

We consider predominately the group $SU(n)$ but with occasional reference to $Sp(r)$, and follow substantially the notation of Corrigan et al. [9], with some minor departures.

It is convenient to define the Euclidean four-vector matrices [2]

$$\alpha_\mu = (-i\sigma, 1), \quad \bar{\alpha}_\mu = (i\sigma, 1), \quad (2.1)$$

which have the basic properties

$$\begin{aligned} \alpha_\mu \bar{\alpha}_\nu &= \delta_{\mu\nu} + i\bar{\sigma}_{\mu\nu}, & \bar{\sigma}_{\mu\nu} &= \bar{\eta}_{\mu\nu}^a \sigma_a, \\ \bar{\alpha}_\mu \alpha_\nu &= \delta_{\mu\nu} + i\sigma_{\mu\nu}, & \sigma_{\mu\nu} &= \eta_{\mu\nu}^a \sigma_a, \\ {}^* \bar{\sigma}_{\mu\nu} &= -\bar{\sigma}_{\mu\nu}, \\ {}^* \sigma_{\mu\nu} &= \sigma_{\mu\nu}, \end{aligned} \quad (2.2)$$

where for an antisymmetric $X_{\mu\nu}$, ${}^* X_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} X_{\alpha\beta}$. The $\alpha_\mu, \bar{\alpha}_\nu$ can be used to construct the Euclidean Dirac matrices as

$$\begin{aligned} \gamma_\mu &= \begin{pmatrix} 0 & \alpha_\mu \\ \bar{\alpha}_\mu & 0 \end{pmatrix}, & \gamma_5 &= \gamma_1 \gamma_2 \gamma_3 \gamma_4 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \\ \{\gamma_\mu, \gamma_\nu\} &= 2\delta_{\mu\nu}, & [\gamma_\mu, \gamma_\nu] &= 2i \begin{pmatrix} \bar{\sigma}_{\mu\nu} & 0 \\ 0 & \sigma_{\mu\nu} \end{pmatrix}. \end{aligned}$$

It is also useful on occasion to display explicit spinorial indices $A, A' = 1, 2$, and write $(\alpha_\mu)_{AA'}, (\bar{\alpha}_\mu)^{A'A}$, where in (2.2) $(\bar{\sigma}_{\mu\nu})_B^A$ and $(\sigma_{\mu\nu})^{B'}_{A'}$ are the generators for $O(4) = SU(2) \times SU(2)$ transformations acting on the unprimed and primed indices, respectively. The antisymmetric tensors $\epsilon^{AB}, \epsilon_{AB}, \epsilon^{A'B'}, \epsilon_{A'B'}$ ($\epsilon^{12} = \epsilon_{12} = 1$) are invariant under separate $SL(2, C)$ transformations on unprimed, primed indices and can be used as a metric, so that

$$(\alpha_\mu)_{AA'} = -\epsilon_{A'B'} (\bar{\alpha}_\mu)^{B'B} \epsilon_{BA}, \quad \epsilon_{AC} \epsilon^{BC} = \delta_A^B. \quad (2.3)$$

The method of constructing a k -instanton solution of the self-dual field equation

$$\begin{aligned} {}^* F_{\mu\nu} &= F_{\mu\nu}, \\ F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu - i[A_\mu, A_\nu], \end{aligned} \quad (2.4)$$

is based on the $(2k + n) \times k$ complex matrices

$$\Delta(x)^A = a^A + b_A \tilde{x}^{A' A}, \quad \tilde{x} = x \cdot \tilde{\alpha}, \quad A = 1, 2. \quad (2.5)$$

The crucial non-linear constraint is in terms of the $k \times k$ matrices

$$\Delta_A^\dagger \Delta^B \propto \delta_A^B, \quad (2.6)$$

for all x , or alternatively with $\Delta(x)_A^\dagger = a_A^\dagger + \tilde{x}_{AA'}^\dagger b^{\dagger A'}$, $\tilde{x}^\dagger = x \cdot \alpha$

$$a_A^\dagger a^B \propto \delta_A^B, \quad b^{\dagger A'} b_{B'} \propto \delta_{B'}^{A'}, \quad b^{\dagger B'} a^B = \epsilon^{B' A'} \epsilon^{BA} b_A a_{A'}^\dagger. \quad (2.7)$$

A given Δ^A satisfying (2.5) may be trivially redefined according to

$$\Delta^A \rightarrow P \Delta^A Q, \quad P \in \text{SU}(2k + n), \quad Q \in \text{GL}(k, \mathbb{C}), \quad (2.8)$$

since as Q includes overall multiplication by a phase it is permissible to restrict $\det P = 1$. The four complex matrices a^A , $b_{A'}$, initially contain $8(2k + n)k$ real parameters but the constraints (2.6), or (2.9), give $10k^2$ conditions so, after allowing for the arbitrariness in (2.8), we have

$$8(2k + n)k - 10k^2 - 2k^2 - (2k + n)^2 + 1 = 4nk - n^2 + 1 \quad (2.9)$$

effective parameters in Δ^A . This result could become negative for k small, n large but this is avoidable since if $n > 2k$ the columns of Δ^A , $A = 1, 2$, can no longer span the full $(2k + n)$ -dimensional complex vector space and instead Δ^A is naturally taken as being a $4k \times k$ matrix. The parameter count, with now $P \in \text{SU}(4k)$, is modified to

$$32k^2 - 10k^2 - 2k^2 - 16k^2 + 1 = 4k^2 + 1. \quad (2.10)$$

The k -instanton gauge field is constructed by considering the solutions of

$$\begin{aligned} \xi^\alpha(x)^\dagger \Delta(x)^A &= 0, \quad \alpha = 1 \dots n, \\ \xi^{\alpha\dagger} \xi^\beta &= \delta^{\alpha\beta}, \end{aligned} \quad (2.11)$$

and then the gauge field is given as an $n \times n$ hermitian generator of $U(n)$,

$$A_\mu^{\alpha\beta}(x) = i \xi^\alpha(x)^\dagger \partial_\mu \xi^\beta(x). \quad (2.12)$$

The ξ^α are defined up to an x -dependent $U(n)$ transformation acting on the indices α , which corresponds to a $U(n)$ gauge transformation on A_μ . The detailed verification of the remarkable result that with A_μ given by the above procedure the field strength is self-dual and has Pontryagin index k is contained in Corrigan et al. [9]. Clearly, from (2.9) and (2.10) the result inherits the full complement of significant parameters in Δ^A , which are invariant under gauge transformation. This parameter count conforms with that found previously [11] and the resultant set of A_μ have been shown to be exhaustive [8].

An important construction in the proof of Corrigan et al. [9], which we shall use

subsequently, is the $(2k + n) \times (2k + n)$ projection operator

$$\begin{aligned} P(x) &= \xi(x) \xi(x)^\dagger = 1 - \Delta(x)^A f(x) \Delta(x)_A^\dagger, \\ \xi(x)^\dagger P(x) &= \xi(x)^\dagger, \quad P^2 = P, \quad \Delta_A^\dagger \Delta^B = f^{-1} \delta_A^B, \end{aligned} \quad (2.13)$$

where we employ the prime constraint (2.6). The Green function for scalar fields belonging to the n -dimensional representation of $U(n)$ in the presence of the gauge field (2.11) is then

$$G(x, x') = \frac{\xi(x)^\dagger \xi(x')}{4\pi^2 (x - x')^2}, \quad (2.14)$$

which satisfies

$$-D_x^2 G(x, x') = \delta(x - x'), \quad D_\mu = \partial_\mu - iA_\mu. \quad (2.15)$$

The gauge group $Sp(r)$ can also be accommodated within this framework by taking $n = 2r$ and imposing the additional reality condition [8]

$$\Delta_A^* = \sigma \epsilon_{AB} \Delta^B,$$

or

$$a_A^* = \epsilon_{AB} \sigma a^B, \quad b^{*A'} = \epsilon^{A'B'} \sigma b_{B'}, \quad (2.16)$$

for $\sigma(2k + r) \times (2k + r)$ with $\sigma^2 = -1$, $\sigma^* = -\sigma^T = \sigma$. It is then possible to require

$$\xi^{\alpha*} = \tilde{\sigma}^{\alpha\beta} \sigma \xi^\beta,$$

with $\tilde{\sigma} 2r \times 2r$ having similar properties to σ above so that the invariance group on the space spanned by ξ^α , $\alpha = 1 \dots 2r$ is now $Sp(r)$ instead of $U(2r)$. In this case a^2 , b_2 are not independent from (2.16) and the non-linear constraints (2.7) become $5k(k - 1)$ conditions on a^1 , b_1 . Since it is now necessary that in (2.8), in order to preserve (2.16), $P \in Sp(k + r)$, $Q \in GL(k, R)$ the parameter count is

$$\begin{aligned} &4(2k + 2r)k - 5k(k - 1) - k^2 - (k + r)(2k + 2r + 1) \\ &= 4(r + 1)k - r(2r + 1), \end{aligned} \quad (2.17)$$

except when $k < r$. Proceeding then as for $U(n)$ the count is

$$16k^2 - 5k(k - 1) - k^2 - 2k(4k + 1) = 2k^2 + 3k. \quad (2.18)$$

This is all in accord with previous results [11]. In Corrigan et al. [9] the reality conditions for this case are automatically incorporated by regarding Δ^A , $A = 1, 2$, as constituting a single $(k + r) \times k$ quaternionic matrix with $\sigma \rightarrow j$.

3. Dirac zero modes

The equations to be solved are

$$\alpha \cdot D \psi^- = 0, \quad \bar{\alpha} \cdot D \psi^+ = 0, \quad D_\mu = \partial_\mu - iA_\mu, \quad (3.1)$$

corresponding to negative, positive chirality solutions of the full Dirac equations. In fact since if $F_{\mu\nu}$ is self-dual, from (2.3)

$$\alpha \cdot D \bar{\alpha} \cdot D = D^2 + \frac{1}{2} \bar{\sigma}_{\mu\nu} F_{\mu\nu} = D^2, \quad (3.2)$$

and as $-D^2$ is a positive definite operator on normalisable functions there are no positive-chirality solutions. The Green functions for $\alpha \cdot D$ can be straightforwardly obtained following Brown et al. [10]:

$$S(x, x') = -\bar{\alpha} \cdot D_x G(x, x'), \quad (3.3)$$

since from (2.15) and (3.2)

$$\alpha \cdot D_x S(x, x') = \delta(x - x'). \quad (3.4)$$

For non-zero Pontryagin number k there are however k normalisable negative-chirality solutions. The Green function for $i\bar{\alpha} \cdot D$ is

$$\bar{S}(x, x') = G(x, x')(-\alpha \cdot \overleftarrow{D}_{x'}), \quad \overleftarrow{D}_\nu = +\overleftarrow{\partial}_\nu + iA_\nu, \quad (3.5)$$

where now

$$-\alpha \cdot D_x \bar{\alpha} \cdot D_x \bar{S}(x, x') = \alpha \cdot D_x \delta(x - x'), \quad (3.6)$$

from (2.15) and (3.2) again. However, instead of (3.4) we must have

$$-\bar{\alpha} \cdot D_x \bar{S}(x, x') = \delta(x - x') - \sum_{i=1}^k \psi_i(x) \psi_i(x')^\dagger, \quad (3.7)$$

displaying the negative-chirality zero mode solutions explicitly. Knowing G in (2.14) then from (3.5) and (3.7) we may find ψ_i by direct calculation following the methods of Corrigan et al. [9]. First we utilise the results that the covariant derivatives act on $\xi^\dagger, \xi$ as follows:

$$D_\mu \xi^\dagger = \xi^\dagger P_\mu + \xi^\dagger \partial_\mu, \quad \xi \overleftarrow{D}_\nu = +P_\nu \xi + \overleftarrow{\partial}_\nu \xi, \quad (3.8)$$

from (2.12) and (2.13), with subscripts denoting differentiation. Then since

$$\bar{\alpha} \cdot \partial_x \alpha \cdot \partial_{x'} \frac{1}{4\pi^2(x - x')^2} = 1\delta(x - x'), \quad \xi(x)^\dagger \xi(x) = 1,$$

we have

$$\begin{aligned} & \bar{\alpha} \cdot D_x G(x, x') \alpha \cdot \overleftarrow{D}_{x'} - 1\delta(x - x') \\ &= \bar{\alpha}_\mu \alpha_\nu \xi(x)^\dagger \left\{ P_\mu(x) P_\nu(x') + 2P_\mu(x) \frac{(x - x')_\nu}{(x - x')^2} \right. \\ & \quad \left. - 2 \frac{(x - x')_\mu}{(x - x')^2} P_\nu(x') \right\} \xi(x') \frac{1}{4\pi^2(x - x')^2}, \end{aligned} \quad (3.9)$$

Since $\xi^\dagger \Delta^A = 0$ from (2.13)

$$\xi^\dagger P_\mu = -\xi^\dagger \Delta_\mu^A f \Delta_A^\dagger, \quad \Delta_\mu^A = b_A \bar{\alpha}_\mu^{A'} A',$$

and with $\Delta(x)_A^\dagger \xi(x') = (\alpha \cdot (x - x'))_{AA'} b^{\dagger A'} \xi(x')$, we obtain

$$\begin{aligned} & \xi(x)^\dagger \{P_\mu(x)(x - x')_\nu - (x - x')_\mu P_\nu(x')\} \xi(x') \\ &= -\xi(x)^\dagger b \{f(x) \bar{\alpha}_\mu \alpha \cdot (x - x')(x - x')_\nu \\ & \quad + (x - x')_\mu \bar{\alpha} \cdot (x - x') \alpha_\nu f(x')\} b^\dagger \xi(x') . \end{aligned} \quad (3.11)$$

Now from (2.2)

$$\begin{aligned} \bar{\alpha}_\mu \alpha_\nu X_{\mu\nu} &= \bar{\alpha}_\mu \alpha_\nu \left\{ \frac{1}{4} \delta_{\mu\nu} X_{\rho\rho} + \frac{1}{2} (X_{[\mu\nu]} + {}^* X_{[\mu\nu]}) \right\} , \\ X_{[\mu\nu]} &= \frac{1}{2} (X_{\mu\nu} - X_{\nu\mu}) , \end{aligned} \quad (3.12)$$

and with

$${}^*(\sigma_{[\mu\sigma} z_\sigma z_\nu]) = -\sigma_{[\mu\sigma} z_\sigma z_\nu] + \frac{1}{2} \sigma_{\mu\nu} z^2 ,$$

we have

$$\begin{aligned} & \bar{\alpha}_\mu \alpha_\nu \xi(x)^\dagger \left\{ P_\mu(x) \frac{(x - x')_\nu}{(x - x')^2} - \frac{(x - x')_\mu}{(x - x')^2} P_\nu(x') \right\} \xi(x') \\ &= -\bar{\alpha}_\mu \alpha_\nu \xi(x)^\dagger b (f(x) + f(x')) \frac{1}{4} \bar{\alpha}_\mu \alpha_\nu b^\dagger \xi(x') . \end{aligned} \quad (3.13)$$

For the remaining term in (3.9)

$$\begin{aligned} & \xi(x)^\dagger P_\mu(x) P_\nu(x') \xi(x') \\ &= \xi(x)^\dagger b \bar{\alpha}_\mu f(x) \Delta(x)^\dagger \Delta(x') f(x') \alpha_\nu b^\dagger \xi(x') , \end{aligned} \quad (3.14)$$

where from the definition of f in (2.13)

$$\begin{aligned} f(x) \Delta(x)_A^\dagger \Delta(x')^B f(x') &= \frac{1}{2} f(x) Z_A^B f(x') + \frac{1}{2} (f(x) + f(x')) \delta_A^B , \\ Z_A^B &= (\Delta(x)_A^\dagger - \Delta(x')_A^\dagger) \Delta(x')^B - \Delta(x)_A^\dagger (\Delta(x)^B - \Delta(x')^B) , \end{aligned} \quad (3.15)$$

so combining (3.5), (3.7), (3.9), (3.13), (3.14) and (3.15):

$$\begin{aligned} & \sum_i \psi_i(x)^{A'} \psi_i(x')_{B'}^\dagger \\ &= -(\bar{\alpha}_\mu \alpha_\nu)^{A'}_{B'} \xi(x)^\dagger b f(x) \bar{\alpha}_\mu Z \alpha_\nu f(x') b^\dagger \xi(x') \frac{1}{8\pi^2 (x - x')^2} . \end{aligned} \quad (3.16)$$

From the construction of Δ we find

$$\begin{aligned} Z &= (\alpha \cdot (x - x') \bar{\alpha}_\rho - \alpha_\rho \bar{\alpha} \cdot (x - x')) M_\rho \\ & \quad + (\alpha \cdot (x - x') \bar{\alpha} \cdot x' - \alpha \cdot x \bar{\alpha} \cdot (x - x')) \nu , \end{aligned} \quad (3.17)$$

where we have employed the constraints (2.7) to write

$$b^{\dagger A'} a^B = (\bar{\alpha}_\rho)^{A'B} M_\rho , \quad a_A^\dagger b_{B'} = (\alpha_\rho)_{AB'} M_\rho , \quad b^{\dagger A'} b_{B'} = \delta_B^{A'} \nu ,$$

with M_ρ , ν $k \times k$ matrices. Inserting (2.19) into (3.16) the terms involving $\bar{\sigma}_{\sigma\rho}$ in the product $\alpha_\sigma \bar{\alpha}_\rho$ give a vanishing contribution from (3.12) using

$$\bar{\alpha}_\mu \bar{\sigma}_{\sigma\rho} \alpha_\mu = 0, \quad \bar{\alpha}_{[\mu} \bar{\sigma}_{\sigma\rho} \alpha_{\nu]} = \bar{\alpha}_{[\sigma} \bar{\sigma}_{\mu\nu} \alpha_{\rho]}, \quad {}^* \bar{\sigma}_{\mu\nu} = -\bar{\sigma}_{\mu\nu}.$$

Hence (3.16) becomes

$$\begin{aligned} \sum_i \psi_i(x)^{A'} \psi_i(x')^{\dagger}_{B'} \\ = \frac{1}{8\pi^2} (\bar{\alpha}_\mu \alpha_\nu)^{A'}{}_{B'} \xi(x)^\dagger b f(x) \bar{\alpha}_\mu \alpha_\nu \nu f(x') b^\dagger \xi(x'), \end{aligned} \quad (3.18)$$

with all singularities as $x \rightarrow x'$ finally cancelled. We may simplify (3.18) with

$$(\bar{\alpha}_\mu)^{A'A} (\bar{\alpha}_\mu)^{C'C} = 2\epsilon^{A'C'} \epsilon^{AC}, \quad \epsilon^{AC} \epsilon_{AC} = 2,$$

so that the eventual result becomes

$$\sum_i \psi_i(x)^{A'} \psi_i(x')^{\dagger}_{B'} = \frac{1}{\pi^2} \epsilon^{A'C'} \epsilon_{B'D'} \xi(x)^\dagger b_{C'} f(x) \nu f(x') b^{\dagger D'} \xi(x'). \quad (3.19)$$

This is obviously of a factorisable form and we can write

$$\psi_i(x)^{A'} = \epsilon^{A'C'} \frac{1}{\pi} (\xi(x)^\dagger b_{C'} f(x))_i, \quad i = 1, \dots, k, \quad (3.20)$$

which has a normalisation, since (3.19) must be the projection operator onto the space of zero mode solutions:

$$\int dx \psi_i(x)^\dagger_{A'} \psi_j(x)^{A'} = (\nu^{-1})_{ij}. \quad (3.21)$$

Clearly there are k independent solutions and Corrigan et al. [9] have shown how, using the arbitrariness in (2.8), $b_{A'}$ can be chosen so that $\nu = 1$. It is also simple to show, using the above methods, that ψ_i solves the Dirac equation (3.1) or, alternatively from (3.20),

$$\begin{aligned} D_\mu \psi'_{iA'} \bar{\alpha}_\mu^{A'A} &= 0, \\ \psi'_{iA'} &= \psi_i^{B'} \epsilon_{B'A'} = \frac{1}{\pi} \xi^\dagger b_{A'} f_i. \end{aligned} \quad (3.22)$$

From the final result (3.20) it is perhaps worth noting that, as a corollary of (3.8) and (3.10), it is possible to express the action of covariant derivatives on $\xi^\dagger$ simply as

$$\begin{aligned} (\bar{\alpha} \cdot D)^{A'A} \xi^\dagger &= 2\pi \psi^{A'} \Delta_B^\dagger \epsilon^{BA}, \\ -D^2 \xi^\dagger &= 4\pi \psi^{A'} \epsilon_{A'B'} b^{\dagger B'}, \end{aligned} \quad (3.23)$$

regarding $\psi_i^{A'}$ as constituting a row vector.

4. 't Hooft solution

In order to obtain an explicit solution of the constraints (2.6) or (2.7) it is necessary to have recourse to restrictive forms for Δ^A which lead eventually to 't Hooft's solution for the $SU(2) = Sp(1)$ case [3]. Letting

$$\Delta(x)^{A'A} = - \begin{bmatrix} \lambda_1 \tilde{x}_0^{A'A} & \dots & \lambda_k \tilde{x}_0^{A'A} \\ \tilde{x}_1^{A'A} & & 0 \\ & \ddots & \\ 0 & & \tilde{x}_k^{A'A} \end{bmatrix}, \quad (4.1)$$

$$x_\lambda = x - x^{(\lambda)}, \quad \tilde{x} = x \cdot \bar{\alpha},$$

where the elements are denoted $\Delta_{\lambda,i}^{A'A}$; $\lambda = 0, \dots, k$; $i = 1, \dots, k$; then A' , λ label the $2k + 2$ rows, i the k columns of Δ^A . It is easy to verify that

$$\Delta_A^\dagger \Delta^B = \delta_A^B (X + x_0^2 \lambda \lambda^T), \quad b^{\dagger A'} b_{B'} = \delta_{B'}^{A'} (1 + \lambda \lambda^T),$$

$$X = \begin{pmatrix} x_1^2 & & 0 \\ & \ddots & \\ 0 & & x_k^2 \end{pmatrix}, \quad \lambda = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_k \end{pmatrix}. \quad (4.2)$$

The solution for two-dimensional $\xi^\dagger$ corresponding to $\Delta(x)$ in (4.1) is written as a $2 \times (2k + 2)$ matrix:

$$\xi(x)^\dagger = \frac{1}{\varphi(x)^{1/2}} \left(\dots c_\lambda \frac{\alpha \cdot x_\lambda}{x_\lambda^2} \dots \right), \quad c_i = -\lambda_i c_0,$$

$$\varphi(x) = \sum_{\lambda=0}^k \frac{c_\lambda^2}{x_\lambda^2}, \quad (4.3)$$

which leads directly to 't Hooft's result (as modified by Jackiw, Nohl and Rebbi [3])

$$A_\mu = -\frac{1}{2\varphi} \bar{\sigma}_{\mu\nu} \partial_\nu \varphi, \quad (4.4)$$

so that $x^{(\lambda)}$ are the positions of the singularities of A_μ . From (4.2)

$$f = X^{-1} - \frac{1}{\varphi} X^{-1} c c^T X^{-1}, \quad c = \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix}, \quad \varphi = c^T X^{-1} c. \quad (4.5)$$

Hence from (4.1) and (4.3) and (4.5) the solution of the Dirac equation from (3.22) becomes

$$\psi'_i(x) = \frac{1}{2\pi} \varphi(x)^{1/2} \alpha \cdot \partial_x \left(\frac{c_i}{x_i^2 \varphi(x)} \right), \quad (4.6)$$

which coincides with the result of Grossman [4]. However, we have an explicit normalisation

$$\int dx \psi_i^\dagger \psi_j = \delta_{ij} - \frac{\lambda_i \lambda_j}{1 + \sum \lambda_i^2} ,$$

from (3.21). In the restricted 't Hooft solution $x^{(0)2} \rightarrow \infty$, $c_0^2/x^{(0)2} \rightarrow 1$, $\lambda_i^2 x^{(0)2} \rightarrow \lambda_i'^2$, $\varphi(x) = 1 + \sum_i \lambda_i'^2/x_i^2$ so that $x^{(i)}$ become the locations of the instantons; the Dirac zero mode solutions are orthogonal.

After this work was completed I received a final copy of ref. [9] in which the authors identify the same solution of the Dirac equation.

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